ASHWIN SHARMA

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PROFESSIONAL SUMMARY

Curious and self-motivated AI enthusiast with a solid foundation in machine learning, data analysis, and statistics. Passionate about applying AI to real-world problems, with hands-on experience in generative AI, NLP, and computer vision. Interested in contributing to innovative teams and building scalable, impactful solutions.

EDUCATION

Bachelor of Technology in Computer Science & Engineering 2025
Amrita Vishwa Vidyapeetham, Coimbatore
Relevant Coursework: Machine Learning, Data Mining, Computer Networks
CGPA: 6.61

Senior Secondary Education (Class XII) 2021
Bhavan's Vidya Mandir, Ernakulam — CGPA : 9.34

Secondary Education (Class X) 2019
Bhavan's Vidya Mandir, Ernakulam — CGPA : 9.24

SKILLS AND INTERESTS

Programming	Python (pandas, numpy, scikit-learn), SQL, TypeScript
ML/AI Tools	TensorFlow, Hugging Face Transformers, LangChain, Google Vertex AI/Gemini, ElevenLabs, Stability AI, Replicate
Web Development	Django, FastAPI, React, Vite, Tailwind CSS
Databases	MySQL, PostgreSQL
Cloud & Deployment	GCP, Vercel, Render, Neon, Gunicorn
Version Control	Git, GitHub
Interests	Artificial Intelligence, Machine Learning, Generative AI, NLP, Data Visualization

EXPERIENCE

Gaude Business & Infrastructure Solutions May 2025 – Present
Associate AI/ML Engineer

- Developed AI-driven pipelines for children's book generation using prompt engineering and multi-modal story.
- Implemented Retrieval-Augmented Generation (RAG) with LangChain to improve reasoning and contextual accuracy.
- Engineered end-to-end workflows automating story creation, illustration generation, and PDF export.
- Designed interactive storytelling chatbot interfaces with customization and safety controls for children.
- Optimized deployment for scalability, enabling concurrent multi-user support with minimal latency.
- Worked on an internal AI-driven system for automated content generation and workflow orchestration using Large Language Models (LLMs).
- Designed and implemented prompt-engineered pipelines for structured text generation with controllable outputs.
- Developed back-end logic for document processing, validation, and automated output generation.
- Improved generation accuracy and consistency by optimizing prompts, chaining strategies, and context handling.

Key Systems Built

MUDU NeoBench – Corporate Learning Platform [***FastAPI, PostgreSQL, React/Vite***]

- Built an organization-based corporate training platform covering course creation, live sessions, learner progress, goals, assessments, notifications, and analytics across learners, trainers, and admin roles.
- Designed the FastAPI backend with async SQLAlchemy, JWT-based auth with role/organization-scoped access control, and Alembic-managed schema migrations.

- Implemented course, quiz, assignment, and goal-tracking modules with grading workflows, submission rules, and event-driven notifications.
- Integrated live-session scheduling via Jitsi and Google Calendar/Meet, plus AI-assisted content and quiz generation hooks using Vertex AI/Gemini.
- Kept AI-generated dashboard and analytics content data-driven rather than model-generated on every load, reducing unnecessary token consumption for routine rendering.
- Built the React + TypeScript frontend with TanStack Query and Zustand, delivering role-specific dashboards for learners, trainers, and organization admins.

Mythora Ecosystem – AI Publishing & Commerce Platform [***Django, React, Vertex AI, ElevenLabs***]

- Built a decoupled dual-application ecosystem combining an AI generative publishing engine with a cultural streaming and e-commerce storefront for Indian mythology content.
- Developed the Core Engine's AI story generation, illustration, activity-book, and audiobook pipelines using Vertex AI/Gemini, Stability AI, Replicate, and ElevenLabs.
- Optimized generation costs by routing image synthesis across Stability AI and Replicate based on cost/quality trade-offs, and by tuning prompt structure and caching to cut redundant Vertex AI/Gemini and ElevenLabs API calls.
- Built the print/PDF fulfillment pipeline generating vectorized, print-ready books with QR-linked companion audio.
- Implemented the Commerce Storefront's video hub, book personalization handshake, and Shloka/Stotram interactive learning with gamified streaks and badges.
- Designed cross-platform integration between the two Django/React apps via session-propagated APIs, and built the e-commerce checkout flow with Razorpay and role-based (superadmin/admin/editor/printer/customer) workflows.

PROJECTS

AuraFit – AI Pantry & Fitness Tracker [***React, Django, PostgreSQL, Gemini AI***]

- Built a mobile-responsive Progressive Web App (PWA) combining AI-powered meal planning, pantry management, and fitness tracking into a single platform.
- Developed full-stack architecture with a React + Vite frontend, Django REST Framework backend, and PostgreSQL (Neon) database.
- Integrated Google Gemini AI to generate personalized meal plans based on pantry contents and nutritional targets.
- Implemented offline-ready app-shell caching and service-worker support for installable, low-latency PWA experience across mobile and desktop.
- Designed and deployed a hybrid cloud architecture with the frontend on Vercel and backend on Render, including API proxying and environment-based configuration.

Carnatic Raga Identification & Kriti Segmentation Annotation [***Python, AI Model Fine-tuning***]

- Built an annotation tool to segment Carnatic kritis into structural sections (pallavi, anu-pallavi, charanam, etc.) and label each segment with its own raga, supporting kritis with raga transitions across segments.
- Designed a data pipeline to combine user-annotated segments with uploaded pre-annotated datasets and external online sources for unified model training.
- Building fine-tuned models for raga identification and in-song raga transition detection using the combined annotated dataset.

Detection of Smart Grey Hole Attack in VANET [***Python***]

- Implemented a hybrid approach combining speed-based clustering and threshold-based classification for detecting grey/black hole attacks in VANET.
- Integrated centralized monitoring to validate local alerts and reduce false positives, achieving improved detection accuracy in simulations.

Gesture Controlled Media [***Python, OpenCV, Mediapipe***]

- Built a real-time hand gesture recognition system to control media playback with play/pause, volume, and track control functions.
- Achieved high responsiveness and low latency with 90%+ recognition accuracy in testing.